

COMPUTER VISION

MASTER CODE CHEAT SHEET

Common model codes + tough/twisted exam templates

Keep this PDF open during the exam • Find the topic • Copy the model • Change parameters

COMPUTER VISION — MODEL CODE CHEAT SHEET

Exam templates: copy the model, then change only the marked parameters/lines. All snippets are designed to be mixed into pipelines.

1. UNIVERSAL STARTERS

<pre># IMAGE LOAD + CHECK import cv2, numpy as np import matplotlib.pyplot as plt img = cv2.imread("image.jpg") if img is None: print("Image not found") gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)</pre>	<pre># SHOW ONE / MANY plt.imshow(rgb); plt.axis("off"); plt.show() plt.figure(figsize=(12,4)) plt.subplot(1,3,1); plt.imshow(gray, cmap="gray"); plt.axis("off") plt.subplot(1,3,2); plt.imshow(img[:, :, :-1]); plt.axis("off") plt.subplot(1,3,3); plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB)); plt.axis("off") plt.show()</pre>	<pre># SAVE / RESIZE / CROP cv2.imwrite("output.jpg", result) small = cv2.resize(img, (500,400)) crop = img[y1:y2, x1:x2] # dimensions h,w = img.shape[:2]</pre>
---	--	---

2. GRAYSCALE • HISTOGRAM • ENHANCEMENT

<pre># HISTOGRAM hist = cv2.calcHist([gray],[0],None,[256],[0,256]) plt.plot(hist); plt.xlim([0,256]); plt.show() # HISTOGRAM EQUALIZATION equalized = cv2.equalizeHist(gray)</pre>	<pre># LOG TRANSFORM g = gray.astype(np.float32) c = 255/np.log(1+np.max(g)) log_img = c*np.log(1+g) log_img = np.uint8(log_img) # gamma <1 brightens; gamma >1 darkens gamma=0.5 gamma_img=np.uint8(255*((gray/255)**gamma))</pre>	<pre># CONTRAST / BRIGHTNESS bright = cv2.convertScaleAbs(img, alpha=1.2, beta=30) # STATS mean=np.mean(gray); std=np.std(gray) mn=np.min(gray); mx=np.max(gray)</pre>
--	--	---

3. THRESHOLDING • BINARY • ROI MASK

<pre># GLOBAL _, binary=cv2.threshold(gray,127,255,cv2.THRESH_BINARY) # OTSU = automatic threshold _, otsu=cv2.threshold(gray,0,255,cv2.THRESH_BINARY+cv2.THRESH_OTSU)</pre>	<pre># ADAPTIVE = uneven lighting adaptive=cv2.adaptiveThreshold(gray,255,cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_BINARY,11,2) # invert if object/background is reversed: cv2.THRESH_BINARY_INV</pre>	<pre># RECTANGULAR ROI roi=gray[y1:y2,x1:x2] # POLYGON ROI mask=np.zeros(gray.shape,np.uint8) pts=np.array([[100,100],[500,100],[500,400],[100,400]],np.int32) cv2.fillPoly(mask,[pts],255) masked=cv2.bitwise_and(gray,gray,mask=mask)</pre>
---	--	--

4. BLUR • EDGES • MORPHOLOGY

<pre># BLUR gauss=cv2.GaussianBlur(gray,(5,5),0) median=cv2.medianBlur(gray,5) # salt/pepper # CANNY edges=cv2.Canny(gauss,50,150)</pre>	<pre># KERNEL + OPENING/CLOSING kernel=np.ones((5,5),np.uint8) opening=cv2.morphologyEx(binary,cv2.MORPH_OPEN,kernel) closing=cv2.morphologyEx(binary,cv2.MORPH_CLOSE,kernel) # opening: remove small noise # closing: fill small gaps/holes</pre>	<pre># ERODE / DILATE eroded=cv2.erode(binary,kernel,iterations=1) dilated=cv2.dilate(binary,kernel,iterations=1) # common pipeline: gray -> blur -> threshold -> morphology -> edges</pre>
---	---	--

5. CONTOURS • OBJECT COUNT • BOUNDING BOX

<pre># FIND CONTOURS contours,_=cv2.findContours(binary,cv2.RETR_EXTERNAL,cv2.CHAIN_APPROX_SIMPLE) for c in contours: area=cv2.contourArea(c) if area>500: x,y,w,h=cv2.boundingRect(c) cv2.rectangle(img,(x,y),(x+w,y+h),(255,0,0),2)</pre>	<pre># COUNT OBJECTS count=0 for c in contours: if cv2.contourArea(c)>500: count+=1 print("Objects:",count) # perimeter peri=cv2.arcLength(c,True)</pre>	<pre># CENTROID M=cv2.moments(c) if M["m00"]!=0: cx=int(M["m10"]/M["m00"]) cy=int(M["m01"]/M["m00"]) cv2.circle(img,(cx,cy),5,(0,0,255),-1)</pre>
--	--	---

6. K-MEANS / COLOR SEGMENTATION

<pre># K-MEANS rgb=cv2.cvtColor(img,cv2.COLOR_BGR2RGB) pixels=np.float32(rgb.reshape((-1,3))) criteria=(cv2.TERM_CRITERIA_EPS+cv2.TERM_CRITERIA_MAX_ITER,100,0.2) K=3 _, labels,centers=cv2.kmeans(pixels,K,None,criteria,10,cv2.KMEANS_RANDOM_CENTERS) centers=np.uint8(centers) seg=centers[labels.flatten()].reshape(rgb.shape)</pre>	<pre># HSV RANGE MASK hsv=cv2.cvtColor(img,cv2.COLOR_BGR2HSV) lower=np.array([H1,S1,V1]) upper=np.array([H2,S2,V2]) mask=cv2.inRange(hsv,lower,upper) result=cv2.bitwise_and(img,img,mask=mask)</pre>	<pre># COLOR SEGMENT + CLEAN kernel=np.ones((5,5),np.uint8) mask=cv2.morphologyEx(mask,cv2.MORPH_OPEN,kernel) mask=cv2.morphologyEx(mask,cv2.MORPH_CLOSE,kernel)</pre>
---	---	--

7. HOUGH LINES • LANES • CIRCLES

<pre># HOUGH LINES edges=cv2.Canny(gray,50,150) lines=cv2.HoughLinesP(edges,1,np.pi/180,threshold=50, minLineLength=50,maxLineGap=20) for line in lines: x1,y1,x2,y2=line[0] cv2.line(img,(x1,y1),(x2,y2),(0,255,0),2)</pre>	<pre># LANE MODEL gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) blur=cv2.GaussianBlur(gray,(5,5),0) edges=cv2.Canny(blur,50,150) roi=cv2.bitwise_and(edges,edges,mask=mask) lines=cv2.HoughLinesP(roi,1,np.pi/180,50, minLineLength=50,maxLineGap=20)</pre>	<pre># HOUGH CIRCLES circles=cv2.HoughCircles(gray,cv2.HOUGH_GRADIENT,dp=1,minDist=30, param1=100,param2=30,minRadius=10,maxRadius=100) if circles is not None: circles=np.uint16(np.around(circles)) for x,y,r in circles[0]: cv2.circle(img,(x,y),r,(0,255,0),2)</pre>
--	---	---

8. AFFINE • ROTATION • TRANSLATION

<pre># ROTATE h,w=img.shape[:2] M=cv2.getRotationMatrix2D((w//2,h//2),30,1.0) rot=cv2.warpAffine(img,M,(w,h))</pre>	<pre># ROTATE + TRANSLATE M=cv2.getRotationMatrix2D((w//2,h//2),30,1.2) M[0,2]+=50 M[1,2]+=30 out=cv2.warpAffine(img,M,(w,h))</pre>	<pre># CUSTOM AFFINE: 3 POINTS src=np.float32([[100,100],[400,100],[100,400]]) dst=np.float32([[150,120],[450,100],[120,450]]) M=cv2.getAffineTransform(src,dst) out=cv2.warpAffine(img,M,(w,h)) # 3 corresponding points -> AFFINE</pre>
---	---	---

9. HOMOGRAPHY • PERSPECTIVE • PANORAMA

<pre># HOMOGRAPHY: 4+ POINTS src=np.float32([[x1,y1],[x2,y2],[x3,y3],[x4,y4]]) dst=np.float32([[a1,b1],[a2,b2],[a3,b3],[a4,b4]]) H,mask=cv2.findHomography(src,dst,cv2.RANSAC,5.0) warped=cv2.warpPerspective(img,H,(w,h)) # 4 corresponding points -> HOMOGRAPHY</pre>	<pre># PERSPECTIVE TRANSFORM corners=np.float32([[0,0],[w,0],[w,h],[0,h]]).reshape(-1,1,2) new=cv2.perspectiveTransform(corners,H) cv2.polylines(out,[np.int32(new)],True,(0,255,0),3)</pre>	<pre># PANORAMA CORE # image1 + image2: SIFT -> BF/FLANN matches -> ratio test -> 4+ good matches -> findHomography(RANSAC) -> warpPerspective -> place/stitch images</pre>
---	---	---

10. SIFT • MATCHING • OBJECT RECOGNITION

<pre>sift=cv2.SIFT_create() kp1,des1=sift.detectAndCompute(gray1,None) kp2,des2=sift.detectAndCompute(gray2,None) bf=cv2.BFMatcher() matches=bf.knnMatch(des1,des2,k=2) good=[] for m,n in matches: if m.distance < 0.75*n.distance: good.append(m)</pre>	<pre># DRAW MATCHES match_img=cv2.drawMatches(img1,kp1,img2,kp2,good,None, flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS) # need >=4 good matches for homography</pre>	<pre># OBJECT RECOGNITION src=np.float32([kp1[m.queryIdx].pt for m in good]).reshape(-1,1,2) dst=np.float32([kp2[m.trainIdx].pt for m in good]).reshape(-1,1,2) H,mask=cv2.findHomography(src,dst,cv2.RANSAC,5.0) inliers=mask.ravel().sum() print("Inliers:",inliers)</pre>
---	---	--

11. IMAGE FUSION • CT/MRI

<pre># SAME SIZE FIRST mri=cv2.resize(mri,(ct.shape[1],ct.shape[0])) # 50/50 fusion fused=cv2.addWeighted(ct,0.5,mri,0.5,0) # 70/30 fused=cv2.addWeighted(ct,0.7,mri,0.3,0)</pre>	<pre># MAX-INTENSITY FUSION img2=cv2.resize(img2,(img1.shape[1],img1.shape[0])) fused=np.maximum(img1,img2) # useful when keeping strongest intensity</pre>	<pre># WEIGHT RULE # output = alpha*A + beta*B + gamma # alpha+beta often = 1 fused=cv2.addWeighted(A,alpha,B,beta,0)</pre>
---	--	---

12. VIDEO • FRAME PROCESSING • MOTION

<pre># READ VIDEO cap=cv2.VideoCapture("video.mp4") while True: ret,frame=cap.read() if not ret: break gray=cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY) edges=cv2.Canny(gray,50,150) cv2.imshow("Edges",edges) if cv2.waitKey(1)&0xFF==ord("q"): break cap.release(); cv2.destroyAllWindows()</pre>	<pre># MOTION BY FRAME DIFFERENCE previous=None # inside loop: gray=cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY) if previous is not None: diff=cv2.absdiff(previous,gray) _,th=cv2.threshold(diff,30,255,cv2.THRESH_BINARY) previous=gray</pre>	<pre># SAVE VIDEO fourcc=cv2.VideoWriter_fourcc(*'mp4v') out=cv2.VideoWriter('out.mp4',fourcc,20, (frame.shape[1],frame.shape[0])) # inside loop: out.write(frame) out.release()</pre>
--	---	--

13. WATERSHED — TOUCHING OBJECTS

<pre>gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) _,binary=cv2.threshold(gray,0,255, cv2.THRESH_BINARY_INV+cv2.THRESH_OTSU) kernel=np.ones((3,3),np.uint8) opening=cv2.morphologyEx(binary,cv2.MORPH_OPEN,kernel,iterations=2) sure_bg=cv2.dilate(opening,kernel,iterations=3)</pre>	<pre>dist=cv2.distanceTransform(opening,cv2.DIST_L2,5) _,sure_fg=cv2.threshold(dist,0.5*dist.max(),255,0) sure_fg=np.uint8(sure_fg) unknown=cv2.subtract(sure_bg,sure_fg) n,markers=cv2.connectedComponents(sure_fg) markers=markers+1 markers[unknown==255]=0</pre>	<pre>markers=cv2.watershed(img,markers) img[markers==-1]=(255,0,0) # idea: # binary -> opening -> background # -> distance transform -> foreground # -> unknown -> connected components # -> watershed</pre>
---	---	--

14. WAVELET • ANOMALY DETECTION

<pre>import pywt coeffs=pywt.dwt2(gray, 'haar') LL, (LH,HL,HH)=coeffs # LL = low frequency # LH/HL/HH = detail/high-frequency bands</pre>	<pre>detail=np.abs(HH) threshold=detail.mean()+2*detail.std() anomaly=(detail>threshold).astype(np.uint8)*255 # adjust 2 -> 3 for stricter detection</pre>	<pre># RECONSTRUCT IF NEEDED reconstructed=pywt.idwt2(((LL, (LH,HL,HH)), 'haar') reconstructed=np.uint8(np.clip(reconstructed,0,255))</pre>
--	---	---

15. SECURITY ZONE • POINT INSIDE POLYGON

<pre># POINT-IN-ZONE zone=np.array([[100,100],[500,100], [500,400],[100,400]],np.int32) point=(x,y) inside=cv2.pointPolygonTest(zone,point,False) if inside>=0: print("INSIDE") else: print("OUTSIDE")</pre>	<pre># DRAW ZONE cv2.polylines(img,[zone],True,(0,255,0),2) cv2.circle(img,point,6,(0,0,255),-1) # test many points: for p in points: if cv2.pointPolygonTest(zone,tuple(p),False)>=0: print("inside:",p)</pre>	<pre># COMBINE WITH MOTION # motion mask + pointPolygonTest # = trigger only when motion is inside # security zone</pre>
--	---	--

16. X-RAY / MEDICAL IMAGE MODEL PIPELINE

<pre># ENHANCEMENT TEMPLATE gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) equalized=cv2.equalizeHist(gray) g=gray.astype(np.float32) c=255/np.log(1+np.max(g)) log_img=np.uint8(c*np.log(1+g)) adaptive=cv2.adaptiveThreshold(equalized,255,cv2.ADAPTIVE_THRESH_MEAN_C, cv2.THRESH_BINARY,11,2)</pre>	<pre># ROI + ENHANCEMENT mask=np.zeros(gray.shape,np.uint8) pts=np.array([[100,100],[500,100], [500,400],[100,400]],np.int32) cv2.fillPoly(mask,[pts],255) roi=cv2.bitwise_and(equalized,equalized,mask=mask)</pre>	<pre># ALWAYS KEEP ORIGINAL IF NEEDED gray = cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) enhanced = cv2.equalizeHist(gray) # use gray for another independent transform # use enhanced for thresholding/segmentation</pre>
--	---	---

17. HIGH-VALUE 'TWIST' TEMPLATES

<pre># FIND ONLY LARGE OBJECTS _,b=cv2.threshold(gray,127,255,cv2.THRESH_BINARY) cs,_=cv2.findContours(b,cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE) large=[c for c in cs if cv2.contourArea(c)>1000]</pre>	<pre># CROP DETECTED BOX x,y,w,h=cv2.boundingRect(c) object_img=img[y:y+h,x:x+w] # filter by aspect ratio aspect=w/h if 0.5<aspect<2.0: print("candidate")</pre>	<pre># EDGE INSIDE ROI roi=cv2.bitwise_and(gray,gray,mask=mask) blur=cv2.GaussianBlur(roi,(5,5),0) edges=cv2.Canny(blur,50,150) # common twist: mask FIRST, then process</pre>
--	---	---

18. PARAMETERS TO CHANGE IN EXAM

<pre>threshold: 127 -> fixed global 0 + OTSU -> automatic 11,2 -> adaptive block/C Canny: 50,150 -> lower/upper edge thresholds Gaussian: (5,5) -> kernel; must be odd</pre>	<pre>HoughLinesP: threshold=50 minLineLength=50 maxLineGap=20 HoughCircles: dp=1, minDist=30 param1=100, param2=30 minRadius/maxRadius</pre>	<pre>SIFT ratio: 0.75 -> stricter if lower RANSAC: 5.0 -> reprojection threshold K-means: K=2/3/4 -> number of clusters Watershed: 0.5*dist.max() -> foreground cutoff</pre>
---	---	--

19. ONE-LINE EXAM MEMORY

<pre>3 corresponding points = AFFINE 4+ corresponding points = HOMOGRAPHY H = warpPerspective Affine M = warpAffine Edges = CANNY Lines = HoughLinesP Circles = HoughCircles Feature matching = SIFT + ratio test</pre>	<pre>Uneven illumination = ADAPTIVE Automatic threshold = OTSU Small noise = OPENING Small holes/gaps = CLOSING Touching objects = WATERSHED Bad matches/outliers = RANSAC ROI application = bitwise_and(..., mask=mask)</pre>	<pre>PANORAMA = SIFT → match → ratio test → Homography → warpPerspective → stitch LANE = gray → blur → Canny → ROI → HoughLinesP OBJECT = threshold → contours → area/box/centroid</pre>
---	--	--

COMPUTER VISION — TOUGHEST EXAM MODEL-CODE CHEAT SHEET

Built as a second sheet for questions that twist/combine lab concepts. Model → modify inputs/parameters. Source topics include feature extraction, convolution, edge detection, segmentation, transformations, wavelets, Hough and SIFT.		
1. MANUAL GRADIENTS — SOBEL / PREWITT / SCHARR		
# SOBEL gray=cv2.imread("image.jpg",0) gx=cv2.Sobel(gray,cv2.CV_64F,1,0,ksize=3) gy=cv2.Sobel(gray,cv2.CV_64F,0,1,ksize=3) mag=cv2.magnitude(gx.astype(np.float32), gy.astype(np.float32)) mag=cv2.convertScaleAbs(mag)	# PREWITT (manual kernels) kx=np.array([[[-1,0,1],[-1,0,1],[-1,0,1]],np.float32]) ky=np.array([[[-1,-1,-1],[0,0,0],[1,1,1]],np.float32]) gx=cv2.filter2D(gray,cv2.CV_32F,kx) gy=cv2.filter2D(gray,cv2.CV_32F,ky) mag=cv2.magnitude(gx,gy) mag=cv2.convertScaleAbs(mag)	# SCHARR gx=cv2.Scharr(gray,cv2.CV_64F,1,0) gy=cv2.Scharr(gray,cv2.CV_64F,0,1) mag=cv2.magnitude(gx.astype(np.float32), gy.astype(np.float32)) mag=cv2.convertScaleAbs(mag)
2. LAPLACIAN / LoG / EDGE COMPARISON		
# LAPLACIAN lap=cv2.Laplacian(gray,cv2.CV_64F) lap=cv2.convertScaleAbs(lap) # second derivative	# LoG = Gaussian then Laplacian blur=cv2.GaussianBlur(gray,(5,5),0) log=cv2.Laplacian(blur,cv2.CV_64F) log=cv2.convertScaleAbs(log)	# EDGE PIPELINE blur=cv2.GaussianBlur(gray,(5,5),0) sobelx=cv2.Sobel(blur,cv2.CV_64F,1,0,3) sobely=cv2.Sobel(blur,cv2.CV_64F,0,1,3) mag=cv2.magnitude(sobelx.astype(np.float32), sobely.astype(np.float32)) edges=cv2.Canny(blur,50,150)
3. CONVOLUTION — MANUAL / KERNEL / OUTPUT SIZE		
# MANUAL 2D CONVOLUTION kernel=np.array([[1,0,-1],[1,0,-1],[1,0,-1]],float) kh,kw=kernel.shape out=np.zeros((gray.shape[0]-kh+1, gray.shape[1]-kw+1)) for i in range(out.shape[0]): for j in range(out.shape[1]): out[i,j]=np.sum(gray[i:i+kh,j:j+kw]*kernel) out=np.uint8(np.clip(out,0,255))	# OPENCV FILTER kernel=np.ones((3,3),np.float32)/9 filtered=cv2.filter2D(gray,-1,kernel) # Gaussian: filtered=cv2.GaussianBlur(gray,(5,5),0)	# OUTPUT SIZE # valid convolution: # out = floor((N-K+2P)/S)+1 N=32; K=3; P=0; S=1 out_size=(N-K+2*P)//S+1 print(out_size) # know: padding P, stride S, kernel K
4. HOG — ACTUAL FEATURE VECTOR TEMPLATE		
from skimage.feature import hog from skimage import color gray=color.rgb2gray(img) features,visual=hog(gray,orientations=9, pixels_per_cell=(8,8), cells_per_block=(2,2), visualize=True) print(features.shape)	# HOG WORKFLOW # image -> gradient magnitude/direction # -> cells -> orientation histograms # -> blocks -> normalized descriptor # # Change: # orientations=9 # pixels_per_cell=(8,8) # cells_per_block=(2,2)	# SHOW HOG plt.figure(figsize=(8,4)) plt.subplot(1,2,1) plt.imshow(gray,cmap="gray"); plt.axis("off") plt.subplot(1,2,2) plt.imshow(visual,cmap="gray"); plt.axis("off") plt.show()
5. LBP — TEXTURE FEATURE TEMPLATE		
from skimage.feature import local_binary_pattern gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) P=8; R=1 lbp=local_binary_pattern(gray,P,R,method="uniform") # histogram of LBP hist,_=np.histogram(lbp.ravel(),bins=np.arange(0,P+3), range=(0,P+2)) hist=hist.astype(float) hist/=hist.sum()	# VISUALIZE plt.imshow(lbp,cmap="gray") plt.axis("off"); plt.show() # P = number of neighbors # R = radius # uniform = common compact pattern	# TOUGH TWIST: # Compare texture in two ROIs a=lbp[y1:y2,x1:x2] b=lbp[y3:y4,x3:x4] ha,_=np.histogram(a.ravel(),bins=np.arange(0,P+3)) hb,_=np.histogram(b.ravel(),bins=np.arange(0,P+3)) print(np.linalg.norm(ha-hb))
6. COLOR HISTOGRAM / HED / HIG-STYLE FEATURES		
# COLOR HISTOGRAM for c in range(3): hist=cv2.calcHist([img],[c],[None],[256],[0,256]) plt.plot(hist) plt.show() # BGR channels; convert RGB if needed	# GRADIENT DIRECTION HISTOGRAM gx=cv2.Sobel(gray,cv2.CV_32F,1,0,3) gy=cv2.Sobel(gray,cv2.CV_32F,0,1,3) mag,ang=cv2.cartToPolar(gx,gy,angleInDegrees=True) hist,_=np.histogram(ang,bins=9,range=(0,360),weights=mag)	# HIG IDEA # HIG = intensity gradients # gradient magnitude weights the direction bins # # HOG/HIG twist: # compute gx, gy -> magnitude + angle # -> histogram -> feature vector
7. REGION GROWING — HARD SEGMENTATION QUESTION		

<pre># SIMPLE REGION GROWING seed=(y0,x0) tol=10 visited=np.zeros(gray.shape,bool) region=np.zeros(gray.shape,np.uint8) queue=[seed] seed_val=int(gray[seed]) while queue: y,x=queue.pop(0) if visited[y,x]: continue visited[y,x]=True if abs(int(gray[y,x])-seed_val)<=tol: region[y,x]=255 for dy,dx in [(-1,0),(1,0),(0,-1),(0,1)]: ny,nx=y+dy,x+dx if 0<=ny<gray.shape[0] and 0<=nx<gray.shape[1]: if not visited[ny,nx]: queue.append((ny,nx))</pre>	<pre># CHANGE CONNECTIVITY # 4-neighbor: [(-1,0),(1,0),(0,-1),(0,1)] # 8-neighbor: [(dy,dx) for dy in (-1,0,1) for dx in (-1,0,1) if (dy,dx)!=(0,0)] # stricter similarity -> smaller tol</pre>	<pre># EXAM LOGIC # seed -> inspect neighbors # -> if similar, add # -> continue from new pixels # -> stop when neighbors fail criterion # seed can be: seed=(100,100) # similarity can use intensity/color distance</pre>
--	--	---

8. WATERSHED — COMPLETE TOUGH TEMPLATE

<pre>gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) _,b=cv2.threshold(gray,0,255, cv2.THRESH_BINARY_INV+cv2.THRESH_OTSU) kernel=np.ones((3,3),np.uint8) opening=cv2.morphologyEx(b,cv2.MORPH_OPEN,kernel,2) sure_bg=cv2.dilate(opening,kernel,3)</pre>	<pre>dist=cv2.distanceTransform(opening,cv2.DIST_L2,5) _,fg=cv2.threshold(dist,0.5*dist.max(),255,0) fg=np.uint8(fg) unknown=cv2.subtract(sure_bg,fg) n,markers=cv2.connectedComponents(fg) markers+=1 markers[unknown==255]=0</pre>	<pre>markers=cv2.watershed(img,markers) img[markers==-1]=(255,0,0) # KEY: # touching objects -> watershed # distance transform -> sure foreground # markers -> separation</pre>
---	---	---

9. ACTIVE CONTOUR / SNAKE — MODEL

<pre># skimage model from skimage.segmentation import active_contour from skimage.filters import gaussian import numpy as np g=gaussian(gray,3) s=np.linspace(0,2*np.pi,100) r=150+80*np.sin(s) c=200+80*np.cos(s) init=np.array([r,c]).T snake=active_contour(g,init,alpha=0.01,beta=10, gamma=0.001)</pre>	<pre># VISUALIZE plt.imshow(gray,cmap="gray") plt.plot(init[:,1],init[:,0],"--r") plt.plot(snake[:,1],snake[:,0],"-b") plt.axis("off"); plt.show() # idea: snake moves toward object boundary # energy = internal + external forces</pre>	<pre># EXAM USE: # if asked "segment an object boundary" # and specifically mentions snakes/active contour: # initialize curve around object # run active_contour # plot final contour</pre>
--	--	--

10. DWT / MULTI-RESOLUTION / ANOMALY

<pre>import pywt LL,(LH,HL,HH)=pywt.dwt2(gray,"haar") plt.figure(figsize=(8,6)) plt.subplot(2,2,1); plt.imshow(LL,cmap="gray") plt.subplot(2,2,2); plt.imshow(LH,cmap="gray") plt.subplot(2,2,3); plt.imshow(HL,cmap="gray") plt.subplot(2,2,4); plt.imshow(HH,cmap="gray") plt.show()</pre>	<pre># ANOMALY FROM HIGH-FREQUENCY DETAIL detail=np.abs(HH) T=detail.mean()+2*detail.std() anom=(detail>T).astype(np.uint8)*255 # increase 2 -> stricter # HH = diagonal detail</pre>	<pre># RECONSTRUCTION rec=pywt.idwt2((LL,(LH,HL,HH)),"haar") rec=np.uint8(np.clip(rec,0,255)) # DWT: # image -> LL + LH + HL + HH # inverse -> reconstruct image</pre>
---	--	---

11. CWT / WPT — KNOW THE CODE PATTERN

<pre># CWT (1D signal) coef,freq=pywt.cwt(signal, scales=np.arange(1,64), wavelet="morl") # coef = time/scale representation # plot: plt.imshow(abs(coef),aspect="auto", cmap="gray")</pre>	<pre># WAVELET PACKET wp=pywt.WaveletPacket(data=gray, wavelet="haar", mode="symmetric",maxlevel=2) for node in wp.get_level(2,"freq"): print(node.path,node.data.shape) # WPT recursively decomposes sub-bands</pre>	<pre># WHEN TO USE DWT -> standard multi-resolution decomposition CWT -> continuous scale/translation analysis WPT -> deeper recursive frequency sub-bands # If exact lab API is not given, # remember the conceptual decomposition.</pre>
---	--	--

12. HOMOGRAPHY FROM POINTS — HARD TWIST

<pre>src=np.float32([[100,100],[500,100], [500,400],[100,400]]) dst=np.float32([[130,80],[520,120], [480,430],[90,390]]) H,status=cv2.findHomography(src,dst,cv2.RANSAC,5.0) warped=cv2.warpPerspective(img,H,(img.shape[1],img.shape[0]))</pre>	<pre># MAP A POINT p=np.float32([[250,200]]) mapped=cv2.perspectiveTransform(p,H) print(mapped) # H maps src plane -> dst plane # if reverse direction is needed: Hinv=np.linalg.inv(H)</pre>	<pre># PERSPECTIVE CORNERS h,w=template.shape[:2] corners=np.float32([[0,0],[w,0],[w,h],[0,h]]).reshape(-1,1,2) mapped=cv2.perspectiveTransform(corners,H) # draw mapped polygon</pre>
--	---	---

13. SIFT + HOMOGRAPHY — FULL OBJECT/PANORAMA QUESTION		
<pre>sift=cv2.SIFT_create() kp1,d1=sift.detectAndCompute(g1,None) kp2,d2=sift.detectAndCompute(g2,None) bf=cv2.BFMatcher() pairs=bf.knnMatch(d1,d2,k=2) good=[m for m,n in pairs if m.distance<0.75*n.distance]</pre>	<pre>src=np.float32([kp1[m.queryIdx].pt for m in good]).reshape(-1,1,2) dst=np.float32([kp2[m.trainIdx].pt for m in good]).reshape(-1,1,2) H,inliers=cv2.findHomography(src,dst,cv2.RANSAC,5.0) print("good:",len(good), "inliers:",int(inliers.sum()))</pre>	<pre># OBJECT CORNERS h,w=img1.shape[:2] corn=np.float32([[0,0],[w,0],[w,h],[0,h]]).reshape(-1,1,2) mapped=cv2.perspectiveTransform(corn,H) cv2.polylines(img2,[np.int32(mapped)], True,(0,255,0),3) # RANSAC removes geometrically inconsistent matches</pre>
14. TOUGHEST INTEGRATED QUESTION — MEDICAL IMAGE		
<pre># Example: # enhance -> denoise -> segment -> measure gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) enh=cv2.equalizeHist(gray) den=cv2.GaussianBlur(enh,(5,5),0) _,b=cv2.threshold(den,0,255, cv2.THRESH_BINARY+cv2.THRESH_OTSU)</pre>	<pre># clean + contours kernel=np.ones((3,3),np.uint8) b=cv2.morphologyEx(b,cv2.MORPH_OPEN,kernel) cs,_=cv2.findContours(b,cv2.RETR_EXTERNAL,cv2.CHAIN_APPROX_SIMPLE) for c in cs: if cv2.contourArea(c)>500: x,y,w,h=cv2.boundingRect(c) print("area=",cv2.contourArea(c))</pre>	<pre># If illumination is uneven: b=cv2.adaptiveThreshold(enh,255,cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_BINARY,11,2) # If objects touch: # replace/simple threshold stage with # watershed pipeline</pre>
15. TOUGHEST INTEGRATED QUESTION — LANE DETECTION		
<pre>gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) blur=cv2.GaussianBlur(gray,(5,5),0) edges=cv2.Canny(blur,50,150) mask=np.zeros(gray.shape,np.uint8) pts=np.array([[100,400],[300,250], [500,250],[700,400]],np.int32) cv2.fillPoly(mask,[pts],255) roi=cv2.bitwise_and(edges,edges,mask=mask)</pre>	<pre>lines=cv2.HoughLinesP(roi,1,np.pi/180,threshold=40, minLineLength=40,maxLineGap=30) for l in lines: x1,y1,x2,y2=l[0] cv2.line(img,(x1,y1),(x2,y2),(0,255,0),3) # twist: change ROI points and # Hough thresholds for the image</pre>	<pre># FULL LOGIC image -> grayscale -> Gaussian blur -> Canny -> polygon ROI -> HoughLinesP -> draw lines # If asked to "ignore sky/side regions": # change polygon mask.</pre>
16. TOUGHEST INTEGRATED QUESTION — SECURITY + MOTION		
<pre># previous + current frame diff=cv2.absdiff(prev,gray) _,motion=cv2.threshold(diff,30,255, cv2.THRESH_BINARY) motion=cv2.morphologyEx(motion,cv2.MORPH_OPEN, np.ones((3,3),np.uint8))</pre>	<pre># detect moving contours cs,_=cv2.findContours(motion,cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE) for c in cs: if cv2.contourArea(c)>500: M=cv2.moments(c) if M["m00"]: p=(int(M["m10"]/M["m00"]), int(M["m01"]/M["m00"]))</pre>	<pre># zone test zone=np.array([[100,100],[500,100], [500,400],[100,400]],np.int32) if cv2.pointPolygonTest(zone,p,False)>=0: print("ALERT: motion inside zone") # motion + contour + centroid + polygon test</pre>
17. EXAM TRAPS — WHAT TO CHANGE		
Question: uneven lighting USE adaptiveThreshold Question: automatic threshold USE OTSU Question: touching objects USE watershed Question: noisy edges Gaussian -> Canny	Question: salt-and-pepper medianBlur Question: 3 point mapping getAffineTransform + warpAffine Question: 4 point perspective findHomography + warpPerspective Question: bad feature matches ratio test + RANSAC	Question: texture LBP Question: shape/object gradients HOG Question: lines HoughLinesP Question: circles HoughCircles Question: multi-resolution DWT
18. LAST-MINUTE HARD-CODE FORMULAS		
AFFINE: x' = A x + b 3 point correspondences warpAffine() HOMOGRAPHY: x' ~ Hx 4+ point correspondences warpPerspective() CONVOLUTION: output=floor((N-K+2P)/S)+1	CANNY: Gaussian -> gradient -> NMS -> double threshold -> hysteresis SIFT: scale-space -> keypoints -> orientation -> descriptor -> matching -> ratio test -> RANSAC homography	SEGMENTATION: threshold / edges / region growing / watershed / K-means / active contour WAVELET: DWT -> LL,LH,HL,HH LL=low-frequency approximation others=detail bands